



Solving constrained optimization problems using a novel genetic algorithm

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ARTICLE INFO

Keywords:

Constrained optimization
Evolutionary algorithms
Genetic algorithms
Genetic operations
Stopping rules

ABSTRACT

A novel genetic algorithm is described in this paper for the problem of constrained optimization. The algorithm incorporates modified genetic operators that preserve the feasibility of the trial solutions encoded in the chromosomes, the stochastic application of a local search procedure and a stopping rule which is based on asymptotic considerations. The algorithm is tested on a series of well-known test problems and a comparison is made against the algorithms C-SOMGA and DONLP2.

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1. Introduction

In general, the constrained optimization problem can be defined as

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p, \end{aligned} \quad (1)$$

where $x_i \in [a_i, b_i]$, $i = 1, \dots, n$. The above problem is applicable in many scientific and practical fields such as physics [1–3], aircraft design [4], astronomy [5], chemistry [6–8], biology [9–11], etc. In order to tackle the above problem many methods have been proposed such as Lagrange multiplier methods [12–14], Trust region methods [15–17], interval methods [18,19], methods that utilize artificial neural networks [20–22], derivative free optimization methods [23–25] etc. Recently, population based stochastic search techniques such as genetic algorithms [26–30], particle swarm optimization [31,32] and differential evolution [33,34] have been used to solve constrained optimization problems. The main advantages of population based methods are

- They do not require the objective function to be differentiable or continuous.
- They do not require the evaluation of gradients.
- They can escape from local minima.

In this article a novel genetic algorithm method for the constrained optimization problem is presented. The proposed method utilizes:

1. Modified versions of the genetic operators of crossover and mutation. These new versions preserve the feasibility of the trial solutions of the constrained problem that are encoded in the chromosomes.

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2. Application of a local search procedure to chromosomes that are selected randomly from the genetic population.
3. A stopping rule based on asymptotic considerations.

The rest of this article is organized as follows: in Section 2 the steps of the proposed method are described in detail, in Section 3 the test functions as well as the experimental results are listed and finally in Section 4 some conclusions are derived and some thoughts for future research are given.

2. Method description

The steps of the proposed method are listed in Algorithm 1. The algorithm initializes a set of K chromosomes inside the bounds $[a, b]$ of the objective function and iteratively applies the modified genetic operators to the chromosomes until some stopping criteria are met. In the following the critical steps of the proposed algorithm are explained in detail.

Algorithm 1. The main steps of the proposed algorithm

1. **Set** the parameters of the algorithm:
 - (a) The number of chromosomes K .
 - (b) The maximum number of generations ITERMAX.
 - (c) The selection rate p_s .
 - (d) The mutation rate p_m .
 - (e) The local search rate p_l .
2. **Set** iter=0.
3. **Initialize** the K chromosomes and store them to the set $X = \{x_1, x_2, \dots, x_K\}$. Every chromosome x_i is initialized randomly inside the feasible region $[a, b]$ of the Eq. (1).
4. **Evaluate** the fitness for every chromosome. This evaluation is performed using a penalty technique as described in Section 2.1.
5. **Apply** the modified genetic operations of crossover and mutation to the population. These operations are described in Section 2.2.
6. **Select** randomly some of the chromosomes from the population and apply to them the local search procedure as described in Section 2.3.
7. **Set** iter=iter+1.
8. **If** the termination criteria are hold **terminate**, **else goto** step 4. The details of the termination criteria are given in Section 2.4.

2.1. Fitness evaluation

The proposed algorithm uses a penalization strategy in order to transform the constrained optimization problem to an unconstrained one. When a trial solution encoded to a chromosome is infeasible, its fitness value is penalized according to the constraint violations. The steps for the evaluation of the fitness for any given chromosome x are the following:

1. Calculate the function value at point x using:

$$v_1(x) = f(x). \quad (2)$$

2. Calculate the violation of the equality constraints $h_i(x), i = 1, \dots, p$ of Eq. (1) as

$$v_2(x) = \sum_{i=1}^p h_i^2(x). \quad (3)$$

3. Calculate the violation of the inequality constraints $g_i(x), i = 1, \dots, m$ of the Eq. (1) as

$$v_3(x) = \sum_{i=1}^m G^2(g_i(x)), \quad (4)$$

where the function $G(x)$ is defined as follows:

$$G(x) = \begin{cases} 0, & x \leq 0, \\ x, & x > 0. \end{cases} \quad (5)$$

4. Compute the fitness value $v(x)$ using the formula

$$v(x) = v_1(x) + \lambda v_2(x) + \lambda v_3(x), \quad (6)$$

where $\lambda > 0$.

2.2. Genetic operations

In every generation, the genetic operations of crossover and mutation are taking place. The process of crossover is presented in Algorithm 2 and the mutation procedure is outlined in Algorithm 3. These genetic operations are the typical genetic operations used in most real coded genetic algorithms, but the feasibility of the chromosomes is preserved. Also, these operations rejects chromosomes that are created outside the bound constraints. The auxiliary procedures $\text{perturb}(x)$ and $\text{perturbFeasible}(x)$ are listed in Algorithms 4 and 5, respectively.

Algorithm 2. The proposed crossover procedure

1. **Sort** the chromosomes $x_1, x_2, \dots, x_K$ with respect to their fitness values $f_1, f_2, \dots, f_K$ in a way such that the best chromosome is placed at the beginning of the population and the worst at the end.
2. **Set** $C = \emptyset$ where C will be the produced children from the crossover procedure.
3. **For** $i = 1, \dots, \frac{(1-p_s) \times K}{2}$ **do**
 - (a) **Select** using tournament selection two chromosomes p_1, p_2 from X .
 - (b) **For** $j = 1..n$ **do**
 - i. **Create** a random number a in the range $[-0.5, 1.5]$.
 - ii. **Set** $c_{1j} = a \times p_{1j} + (1 - a)p_{2j}$
 - iii. **Set** $c_{2j} = a \times p_{2j} + (1 - a)p_{1j}$
 - (c) **End for**
 - (d) $C = C \cup c_1$
 - (e) $C = C \cup c_2$
4. **End for**
5. **For** $i = 1..(1 - p_s) \times K$ **do**
 - (a) **Take** the next element c_i from the set of children C .
 - (b) **If** (x_{K-i-1} is feasible and c_i is feasible) **or** (x_{K-i-1} is not feasible and c_i is not feasible) **then** $x_{K-i-1} = c_i$
6. **End for**

Algorithm 3. The proposed mutation procedure

1. **For** $i = 2..K$ **do**
 - (a) **If** x_i is feasible **then** $\text{perturbFeasible}(x_i)$
 - (b) **else** $\text{perturb}(x_i)$
 - (c) **endif**
2. **End for**

Algorithm 4. The procedure $\text{perturb}(x)$

1. **For** $j = 1, \dots, n$ **do**
 - (a) **Let** r a random number in $(0, 1)$
 - (b) **If** $r \leq p_m$ **then** $x_j = a_j + \xi \times (b_j - a_j)$, where ξ a random number in $(0, 1)$.
 - (c) **endif**
2. **End for**

Algorithm 5. The procedure $\text{perturbFeasible}(x)$

1. **For** $j = 1..n$ **do**
2. **Let** r a random number in $(0, 1)$
 - (a) **If** $r \leq p_m$ **then**
 - i. **Set** $\text{tries} = 0$
 - ii. **Set** $x_{\text{old}} = x_j$
 - iii. **Set** $x_j = a_j + \xi \times (b_j - a_j)$, where ξ a random number in $(0, 1)$.
 - iv. **If** (x is not feasible) **then** $x_j = x_{\text{old}}$
 - v. $\text{tries} = \text{tries} + 1$
 - vi. **If** $\text{tries} \leq \text{MAXTRIES}$ **goto** step 2.a.ii
 - (b) **endif**
3. **End for**

2.3. Application of local search

In every generation the application of a local search procedure to some randomly selected chromosomes is taking place. The purpose of this applications is to improve the fitness of the chromosomes and to speed up the convergence of the algorithm to the global minimum. The steps of this procedure are the following:

1. **For** $i = 1, \dots, K$ **do**
 - (a) **Let** r a random number in $(0, 1)$.
 - (b) **If** $r \leq p_l$ **then**
 - (i.) **Apply** a local search procedure to the chromosome x_i
 - (c) **endif**
2. **End for**

The used local search procedure is the algorithm Cobyla due to Powell [35].

2.4. Termination criteria

In this paper a stopping rule based on asymptotic considerations is used. The algorithm records in every generation the variance of the best discovered value. If there is not any improvement for a number of generations we can assume that the algorithm have managed to discover the global minimum and hence the algorithm should terminate. This stopping rule was proposed recently in the paper of Tsoulos [36] and it has as follows:

1. At every generation iter, the variance $\sigma^{(\text{iter})}$ of the best discovered value is calculated.
2. The genetic algorithm terminates when

$$\sigma^{(\text{iter})} \leq \frac{\sigma^{(\text{last})}}{2} \quad \text{OR} \quad \text{iter} > \text{ITERMAX}, \quad (7)$$

where last is the generation where the current best value was discovered for the first time.

3. Experiments

The proposed method was tested on a series of well-known test problems from the relevant literature. In the following, these test problems are described followed by the experimental results of the proposed method.

3.1. Test functions

Levy

This problem is described in [37] and it is given by

$$\min_x f(x) = -x_1 - x_2,$$

with $x \in [0, 1]^2$, subject to

$$g_1(x) = \left[(x_1 - 1)^2 + (x_2 - 1) \right] \left(\frac{1}{2a^2} - \frac{1}{2b^2} \right) + (x_1 - 1)(x_2 - 1) \left(\frac{1}{a^2} - \frac{1}{b^2} \right) - 1 \geq 0,$$

with $a = 2, b = 0.25$. The value of global minimum is $f_{\min} = -1.8729$.

Salkin

This problem is described in [38] and it is given by

$$\max_x f(x) = 3x_1 + x_2 + 2x_3 + x_4 - x_5,$$

with $1 \leq x_1 \leq 4, 80 \leq x_2 \leq 88, 30 \leq x_3 \leq 35, 145 \leq x_4 \leq 150, 0 \leq x_5 \leq 2$ subject to

$$g_1(x) = 25x_1 - 40x_2 + 16x_3 + 21x_4 + x_5 \leq 300,$$

$$g_2(x) = x_1 + 20x_2 - 50x_3 + x_4 - x_5 \leq 200,$$

$$g_3(x) = 60x_1 + x_2 - x_3 + 2x_4 + x_5 \leq 600,$$

$$g_4(x) = -7x_1 + 4x_2 + 15x_3 - x_4 + 65x_5 \leq 700.$$

This global maximum is $f_{\max} = 320$.

Himmelblau

This problem is described in [39] and it is given by

$$\min_x f(x) = 4.3x_1 + 31.8x_2 + 63.3x_3 + 15.8x_4 + 68.5x_5 + 4.7x_6,$$

with $0 \leq x_1 \leq 0.31$, $0 \leq x_2 \leq 0.046$, $0 \leq x_3 \leq 0.068$, $0 \leq x_4 \leq 0.042$, $0 \leq x_5 \leq 0.028$, $0 \leq x_6 \leq 0.0134$ subject to

$$g_1(x) = 17.1x_1 + 38.2x_2 + 204.2x_3 + 212.3x_4 + 623.4x_5 + 1495.5x_6 - 169x_1x_3 - 3580x_3x_5 \\ - 3810x_4x_5 - 18500x_4x_6 - 24300x_5x_6 - 4.97 \geq 0,$$

$$g_2(x) = 1.88 + 17.9x_1 + 36.8x_2 + 113.9x_3 + 169.7x_4 + 337.8x_5 + 1385.2x_6 - 139x_1x_3 - 2450x_4x_5 \\ - 600x_4x_6 - 17200x_5x_6 \geq 0,$$

$$g_3(x) = 429.08 - 273x_2 - 70x_4 - 819x_5 + 26000x_4x_5 \geq 0,$$

$$g_4(x) = 159.9x_1 - 311x_2 + 587x_4 + 391x_5 + 2198x_6 - 14000x_1x_6 + 78.02 \geq 0.$$

The value of global minimum is $f_{\min} = 0.0156$.

Hess

This problem is described in [40] and it is given by

$$\max_x f(x) = 25(x_1 - 2)^2 + (x_2 - 2)^2 + (x_3 - 1)^2 + (x_4 - 4)^2 + (x_5 - 1)^2 + (x_6 - 4)^2,$$

with $0 \leq x_1 \leq 5$, $0 \leq x_2 \leq 1$, $1 \leq x_3 \leq 5$, $0 \leq x_4 \leq 6$, $0 \leq x_5 \leq 5$, $0 \leq x_6 \leq 10$ subject to

$$g_1(x) = x_1 + x_2 - 2 \geq 0,$$

$$g_2(x) = -x_1 + x_2 + 6 \geq 0,$$

$$g_3(x) = x_1 - x_2 + 2 \geq 0,$$

$$g_4(x) = -x_1 + 3x_2 + 2 \geq 0,$$

$$g_5(x) = (x_3 - 3)^2 + x_4 - 4 \geq 0,$$

$$g_6(x) = (x_5 - 3)^2 + x_6 - 4 \geq 0.$$

The value of global maximum is $f_{\max} = 310$.

SHITTOKOWSKI

The problem is described in [41] and it is given by

$$\min_x f(x) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2,$$

with $x \in [0, 6]^2$ subject to

$$g_1(x) = 4.84 - (x_1 - 0.05)^2 - (x_2 - 2.5)^2 \geq 0,$$

$$g_2(x) = x_1^2 + (x_2 - 2.5)^2 - 4.84 \geq 0.$$

The value of global minimum is $f_{\min} = 13.59085$.

CHOOTINANI

This problem is described in [42] and it is given by

$$\min_x f(x) = 5 \sum_{i=1}^4 x_i - 5 \sum_{i=1}^4 x_i^2 - \sum_{i=1}^{13} x_i,$$

with $0 \leq x_i \leq 1$ for $i = 1, \dots, 9, 13$, $0 \leq x_i \leq 100$ for $i = 10, 11, 12$ with the following constraints:

$$g_1(x) = 10 - (2x_1 + 2x_2 + x_{10} + x_{11}) \geq 0,$$

$$g_2(x) = 10 - (2x_1 + 2x_3 + x_{10} + x_{12}) \geq 0,$$

$$g_3(x) = 10 - (2x_2 + 2x_3 + x_{11} + x_{12}) \geq 0,$$

$$g_4(x) = 8x_1 - x_{10} \geq 0,$$

$$g_5(x) = 8x_2 - x_{11} \geq 0,$$

$$g_6(x) = 8x_3 - x_{12} \geq 0,$$

$$g_7(x) = 2x_4 + x_5 - x_{10} \geq 0,$$

$$g_8(x) = 2x_6 + x_7 - x_{11} \geq 0,$$

$$g_9(x) = 2x_8 + x_9 - x_{12} \geq 0.$$

The value of global minimum is $f_{\min} = -15.0$.

CHOOTINAN2

The source of this problem is [42] and it is given by

$$\max_x f(x) = \frac{\sin^3(2\pi x_1) \sin(2\pi x_2)}{x_1^3(x_1 + x_2)},$$

with $x \in [0, 10]^2$. The constraints are given by

$$\begin{aligned} g_1(x) &= -x_1^2 + x_2 - 1 \geq 0, \\ g_2(x) &= -1 + x_1 - (x_2 - 4)^2 \geq 0. \end{aligned}$$

The value of global maximum is $f_{\max} = 0.095$.

CHOOTINAN3

This problem is described in [42] and it is given by

$$\min_x f(x) = x_1^2 + (x_2 - 1)^2,$$

with $x \in [-1, 1]^2$. The constraints are given by

$$h_1(x) = x_2 - x_1^2 = 0.$$

The value of global minimum is $f_{\min} = 0.75$.

CHOOTINAN4

The source of this problem is [42] and it is given by

$$\max_x f(x) = (\sqrt{n})^n \prod_{i=1}^n x_i,$$

with $x \in [0, 1]^n$. The constraints are given by

$$h_1(x) = \sum_{i=1}^n x_i^2 - 1 = 0.$$

In this paper we used as n the values $n = 2, 10, 20, 30, 40$ with names CHOOTINAN4(2), CHOOTINAN4(10), CHOOTINAN4(20), CHOOTINAN4(30) and CHOOTINAN4(40).

CHOOTINAN5

This problem is described in [42] and it is given by

$$\max_x f(x) = \left| \frac{\sum_{i=1}^n \cos^4(x_i) - 2 \prod_{i=1}^n \cos^2(x_i)}{\sqrt{\sum_{i=1}^n i x_i^2}} \right|$$

with $x \in [0, 10]^n$ subject to

$$\begin{aligned} g_1(x) &= \prod_{i=1}^n x_i - 0.75 \geq 0, \\ g_2(x) &= 7.5n - \sum_{i=1}^n x_i \geq 0. \end{aligned}$$

In this paper we used the value of $n = 10, 20, 30, 40$ with names CHOOTINAN5(10), CHOOTINAN5(20), CHOOTINAN5(30) and CHOOTINAN5(40).

LIN1

This problem is described in [43] and it is given by

$$\min_x f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2,$$

with $-0.5 \leq x_1 \leq 0.5$, $x_2 \leq 10$. The constraints are given by

$$\begin{aligned} g_1(x) &= x_1 + x_2^2 \geq 0, \\ g_2(x) &= x_1^2 + x_2 \geq 0. \end{aligned}$$

The value of global minimum is $f_{\min} = 0.25000$.

LIN2

This problem is described in [43] and it is given by

$$\min_x f(x) = -x_1 - x_2,$$

with $0 \leq x_1 \leq 3$, $0 \leq x_2 \leq 4$ subject to

$$g_1(x) = 2x_1^4 - 8x_1^3 + 8x_1^2 - x_2 + 2 \geq 0,$$

$$g_2(x) = 4x_1^4 - 32x_1^3 + 88x_1^2 - 96x_1 - x_2 + 36 \geq 0.$$

The value of global minimum is $f_{\min} = -5.5081$.

G15

This problem is described in [44] and it is given by

$$\min_x f(x) = 1000 - x_1^2 - 2x_2^2 - x_3^2 - x_1x_2 - x_1x_3,$$

with $x \in [0, 10]^3$ subject to the following constraints:

$$h_1(x) = x_1^2 + x_2^2 + x_3^2 - 25 = 0,$$

$$h_2(x) = 8x_1 + 14x_2 + 7x_3 - 56 = 0.$$

The value of the global minimum is $f_{\min} = 961.7150$.

LIN3

This problem is described in [43] and it is given by

$$\min_x f(x) = 0.01x_1^2 + x_2^2,$$

with $2 \leq x_1 \leq 50$, $0 \leq x_2 \leq 50$ subject to

$$g_1(x) = x_1x_2 - 25 \geq 0,$$

$$g_2(x) = x_1^2 + x_2^2 - 25 \geq 0.$$

The value of global minimum is $f_{\min} = 5.0000$

BEAM

The welded beam design problem is described in [45] and it is given by

$$\min_x f(x) = 1.10471x_1^2x_2 + 0.04811x_3x_4(14 + x_2),$$

with $0.1 \leq x_1, x_4 \leq 2$, $0.1 \leq x_2, x_3 \leq 10$ subject to

$$g_1(x) = \tau(x) - 13600 \leq 0,$$

$$g_2(x) = \sigma(x) - 20000 \leq 0,$$

$$g_3(x) = x_1 - x_4 \leq 0,$$

$$g_4(x) = 0.10471x_1^2 + 0.04811x_3x_4(14 + x_2) - 5 \leq 0,$$

$$g_5(x) = 0.125 - x_1 \leq 0,$$

$$g_6(x) = \delta(x) - 0.25 \leq 0,$$

$$g_7(x) = P - P_c(x) \leq 0,$$

where

$$\tau(x) = \sqrt{(\tau')^2 + 2\tau'\tau''\frac{x_2}{2R} + (\tau'')^2},$$

$$\tau' = \frac{P}{\sqrt{2}x_1x_2},$$

$$\tau'' = \frac{QR}{J},$$

$$Q = P\left(L + \frac{x_2}{2}\right),$$

$$R = \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1 + x_3}{2}\right)^2},$$

$$J = 2 \left(\sqrt{2} x_1 x_2 \left(\frac{x_2^2}{12} + \left(\frac{x_1 + x_3}{2}\right)^2 \right) \right),$$

$$\sigma(x) = \frac{6PL}{x_4 x_3^2},$$

$$\delta(x) = \frac{4PL^3}{Ex_3^3 x_4},$$

$$P_c(x) = \frac{4.013E \sqrt{\frac{x_2^2 x_4^6}{36}}}{L_2} \left(1 - \frac{x_3}{2L} \sqrt{\frac{E}{4G}} \right),$$

$$P = 600, \quad L = 14, \quad E = 30 \times 10^6, \quad G = 12 \times 10^6.$$

The value of global minimum is $f_{\min} = 1.724852$.

3.2. Experimental results

The proposed method was tested on the series of test problems described previously using the parameters given in Table 1. The method was tested against the C-SOMGA described in [26] and the method DONLP2 available from the Internet that is based on [46,47]. In Table 2 the results from the application of the proposed method to the test problems are listed. In Table 3 the reported results from the application of method C-SOMGA to the test problems and in Table 4 the experimental results from the application of the method DONLP2 are reported. The proposed method as well as the DONLP2 was applied 100 times on every test problem using different seed for the random number generator each time. The columns in tables have the following meaning:

Table 1

Parameters of the algorithm.

Parameter	Value
K	200
ITERMAX	200
p_s	0.1
p_m	0.05
p_l	0.01
λ	10^3
MAXTRIES	10

Table 2

Experimental results with the proposed method.

Problem	FEVALS	BEST	MEAN	STD
LEVY	4572	−1.8730	−1.8730	4.82×10^{-6}
SALKIN	7244	320.0000	320.0000	0.0
HIMMELBLAU	23539	0.01561	0.01563	5.71×10^{-5}
HESS	16139	310.0000	309.9999	4.5×10^{-4}
SHITTKOWSKI	17483	13.5907	13.5937	6.61×10^{-3}
CHOOTINAN1	21833	−15.0000	−14.9999	7.59×10^{-5}
CHOOTINAN2	3147	0.09582	0.09582	9.81×10^{-10}
CHOOTINAN3	4651	0.7500	0.75003	1.11×10^{-6}
CHOOTINAN4(2)	4940	1.0000	1.0001	6.71×10^{-5}
CHOOTINAN4(10)	15240	1.0125	1.0124	3.47×10^{-5}
CHOOTINAN4(20)	31118	1.0319	1.0318	1.8×10^{-4}
CHOOTINAN4(30)	45238	1.0482	1.0477	3.66×10^{-4}
CHOOTINAN4(40)	91778	1.0645	1.0636	6.9×10^{-4}
CHOOTINAN5(10)	183025	0.7473	0.7118	3.30×10^{-2}
CHOOTINAN5(20)	483870	0.79466	0.7555	2.71×10^{-2}
CHOOTINAN5(30)	658320	0.8078	0.7845	1.73×10^{-2}
CHOOTINAN5(40)	753240	0.7692	0.7219	3.8×10^{-2}
LIN1	5633	0.2500	0.2500	6.78×10^{-7}
LIN2	2431	−5.5080	−5.5080	2.17×10^{-6}
G15	3593	961.71515	961.71516	1.88×10^{-5}
LIN3	15516	5.0000	5.0010	2.78×10^{-3}
BEAM	5608	1.725934	1.725937	3.30×10^{-5}

Table 3

Experimental results with the method CSOMGA.

PROBLEM	MEAN	STD
LEVY	−1.86839	0.00398
SALKIN	319.99270	0.00229
HIMMELBLAU	0.01531	0.00026
HESS	309.15	3.08913
SHITTKOWSKI	13.59610	0.00315
CHOOTINAN1	−14.99225	0.00304
CHOOTINAN2	0.08816	0.01657
CHOOTINAN3	0.82519	0.09761
CHOOTINAN4(2)	0.88794	0.21215
CHOOTINAN5(20)	0.77542	0.02739

Table 4

Experimental results with the DONLP2 method.

PROBLEM	FEVALS	MEAN	STD
LEVY	84	−1.8730	1.2×10^{-7}
SALKIN	281	319.7423	0.7758
HIMMELBLAU	187	1.3880	2.1793
HESS	343	262.6723	55.4475
SHITTKOWSKI	437	18.6812	35.6322
CHOOTINAN1	184	−11.3781	0.4190
CHOOTINAN2	165	0.03687	0.0243
CHOOTINAN3	122	0.75	7.2×10^{-8}
CHOOTINAN4(2)	97	1.00	9.6×10^{-8}
CHOOTINAN4(10)	2008	0.99	0.0995
CHOOTINAN4(20)	5388	0.96	0.1960
CHOOTINAN4(30)	11577	0.97	0.1706
CHOOTINAN4(40)	18484	0.86	0.3470
CHOOTINAN5(10)	3743	0.2249	0.0687
CHOOTINAN5(20)	22863	0.2297	0.0551
CHOOTINAN5(30)	63267	0.2254	0.0396
CHOOTINAN5(40)	43400	0.2209	0.0370
LIN1	138	0.25	0
LIN2	78	−4.5414	0.9183
G15	139	875.1608	275.2253
LIN3	248	5.00	1.7×10^{-6}
BEAM	28851	2.1177	0.7417

1. The column PROBLEM denotes the name of the test problem as given in the previous section.
2. The column BEST denotes the best value located by the proposed algorithm.
3. The column MEAN denotes the mean of the optimal objective function values, located by the proposed algorithm.
4. The column STD denotes the standard deviation of the optimal objective function values.
5. The column FEVALS denotes the average number of function evaluations spent by the proposed algorithm.

As we can see, in most cases where comparative results are available, the proposed method overcomes C-SOMGA in mean and standard deviation values. Also, the proposed method has managed to discover the global minimum in almost every objective function. On the other hand, the method DONLP2 overcomes the proposed method in function evaluations for

Table 5Experiments with the parameter λ .

PROBLEM	$\lambda = 10$	$\lambda = 10^2$	$\lambda = 10^3$	$\lambda = 10^4$
LEVY	4618	4667	4572	4714
SALKIN	7185	7198	7244	7402
HIMMELBLAU	25048	25054	23539	25285
HESS	17201	16983	16139	17113
SHITTKOWSKI	19107	19465	17483	17279
LIN1	5986	5815	5633	8204
LIN2	2396	2314	2431	2350
G15	3860	3755	3593	3860
LIN3	16233	16107	15516	15845
BEAM	5856	5782	5608	5963

the five problems LEVY, CHOOTINAN3, CHOOTINAN4(2), LIN1 and LIN3 but the proposed method is better for the rest of the problems in mean and standard deviation values.

3.3. Experiments with the parameter λ

The choice of the parameter λ is very crucial for the algorithm, because small values can lead to search outside the feasible region and produce infeasible solutions, while greater penalties can make difficult or even impossible the search for optimum solutions. In Table 5 the experiments with various values for the parameter λ for some of the test problems are listed. The numbers in the cells represents the average number of function evaluations from 100 runs using different seed for the random generator each time. It is obvious from the experimental results that the choice of the parameter λ has a small effect to the number of function evaluations required in order to discover the global minimum.

4. Conclusions

A novel genetic algorithm for the location of the global minimum of constrained optimization problems was presented in this paper. The method suggested (a) modified genetic operators that preserve the feasibility of the chromosomes (b) applications of a local search procedure to randomly selected chromosomes and (c) a stopping rule based on asymptotic considerations. The method was tested on a series of well-known of test functions and proved to be successful in the majority of test problems. Moreover, the proposed method can be implemented very easily and the utilization of the proposed stopping significantly improves the speed of the method. Future research will include the use of more efficient initialization techniques as well as the use of other local search procedures.

References

- [1] O.A. Sauer, D.M. Shepard, T.R. Mackie, Application of constrained optimization to radiotherapy planning, *Medical Physics* 26 (1999) 2359–2366.
- [2] E.G. Birgin, I. Chambouleyron, J.M. Martínez, Estimation of the optical constants and the thickness of thin films using unconstrained optimization, *Journal of Computational Physics* 151 (1999) 862–880.
- [3] C. Yang, J.C. Meza, L.W. Wang, A constrained optimization algorithm for total energy minimization in electronic structure calculations, *Journal of Computational Physics* 217 (2006) 709–721.
- [4] K.D. Lee, S. Eyi, Transonic airfoil design by constrained optimization, *Journal of Aircraft* 30 (1993) 805–806.
- [5] F.P. Seelos, R.E. Arvidson, Bounded variable least squares – application of a constrained optimization algorithm to the analysis of TES Emissivity Spectra, in: 34th Annual Lunar and Planetary Science Conference, March 17–21, 2003, League City, Texas (abstract no. 1817).
- [6] M.J. Field, Constrained optimization of ab initio and semiempirical Hartree–Fock wave functions using direct minimization or simulated annealing, *Journal of Physical Chemistry* 95 (1991) 5104–5108.
- [7] G.A. Williams, J.M. Dugan, R.B. Altman, Constrained global optimization for estimating molecular structure from atomic distances, *Journal of Computational Biology* 8 (2001) 523–547.
- [8] J. Baker, D. Bergeron, Constrained optimization in Cartesian coordinates, *Journal of Computational Chemistry* 14 (1993) 1339–1346.
- [9] J.E.A. Bertram, A. Ruina, Multiple walking speed–frequency relations are predicted by constrained optimization, *Journal of Theoretical Biology* 209 (2001) 445–453.
- [10] J.E.A. Bertram, Constrained optimization in human walking: cost minimization and gait plasticity, *Journal of Experimental Biology* 208 (2005) 979–991.
- [11] I. Iakovidis, R.M. Gulrajani, Regularization of the inverse epicardial solution using linearly constrained optimization, in: *Engineering in Medicine and Biology Society, Proceedings of the Annual International Conference of the IEEE Publication Date: 31 October–3 November, 1991*, vol. 13, pp. 698–699.
- [12] D.P. Bertsekas, *Constrained Optimization and Lagrange Multiplier Methods*, Academic Press, New York, 1982.
- [13] D.P. Bertsekas, A.E. Ozdaglar, Pseudonormality and a Lagrange multiplier theory for constrained optimization, *Journal of Optimization Theory and Applications* 114 (2004) 287–343.
- [14] P.E. Gill, W. Murray, The computation of Lagrange-multiplier estimates for constrained minimization, *Mathematical Programming* 17 (1979) 32–60.
- [15] M.J.D. Powell, Y. Yuan, A trust region algorithm for equality constrained optimization, *Mathematical Programming* 49 (2005) 189–211.
- [16] R.H. Byrd, R.B. Schnabel, G.A. Shultz, A trust region algorithm for nonlinearly constrained optimization, *SIAM Journal on Numerical Analysis* 24 (1987) 1152–1170.
- [17] D.M. Gay, A trust-region approach to linearly constrained optimization, *Lecture Notes in Mathematics*, vol. 1066, 1984, pp. 72–105 (Áññáíóéí (ñ)).
- [18] M.Cs. Markót, J. Fernández, L.G. Casado, T. Csendes, New interval methods for constrained global optimization, *Mathematical Programming* 106 (2005) 287–318.
- [19] K. Ichida, Constrained optimization using interval analysis, *Computers and Industrial Engineering* 31 (1996) 933–937.
- [20] W.E. Lillo, S. Hui, S.H. Zak, Neural networks for constrained optimization problems, *International Journal of Circuit Theory and Applications* 21 (1993) 385–399.
- [21] W.E. Lillo, M.H. Loh, S. Hui, S.H. Zak, On solving constrained optimization problems with neural networks: a penalty method approach, *IEEE Transactions on Neural Networks* 4 (1993) 931–940.
- [22] S. Zhang, X. Zhu, L.H. Zou, Second-order neural nets for constrained optimization, *IEEE Transactions on Neural Networks* 3 (1992) 1021–1024.
- [23] S. Lucidi, M. Sciandrone, P. Tseng, Objective-derivative-free methods for constrained optimization, *Mathematical Programming* 92 (1999) 37–59.
- [24] G. Liuzzi, S. Lucidi, A derivative-free algorithm for nonlinear programming, TR 17/05, Department of Computer and Systems Science, Antonio Ruberti, University of Rome, La Sapienza, 2005.
- [25] M.B. Subrahmanyam, An extension of the simplex method to constrained nonlinear optimization, *Journal of Optimization Theory and Applications* 62 (1989) 311–319.
- [26] K. Deep, Dipti, A self-organizing migrating genetic algorithm for constrained optimization, *Applied Mathematics and Computation* 198 (2008) 237–250.
- [27] A. Homaifar, Constrained optimization via genetic algorithms, *Simulation* 62 (1994) 242–253.
- [28] S. Venkatraman, G.G. Yen, A generic framework for constrained optimization using genetic algorithms, *IEEE Transactions on Evolutionary Computation* 9 (2005) 424–435.
- [29] Z. Michalewicz, M. Schoenauer, Evolutionary algorithms for constrained parameter optimization problems, *Evolutionary Computation* 4 (1996) 1–32.
- [30] V.S. Summanwar, V.K. Jayaraman, B.D. Kulkarni, H.S. Kusumakar, K. Gupta, J. Rajesh, Solution of constrained optimization problems by multi-objective genetic algorithm, *Computers and Chemical Engineering* 26 (2002) 1481–1492.

- [31] Q. He, L. Wang, A hybrid particle swarm optimization with a feasibility – based rule for constrained optimization, *Applied Mathematics and Computation* 186 (2007) 1407–1422.
- [32] X.H. Hu, R.C. Eberhart, Solving constrained nonlinear optimization problems with particle swarm optimization, in: N. Callaos (Ed.), *Proceedings of the Sixth World Multiconference on Systematics, Cybergenetics and Informatics*, Orlando, FL, 2002, pp. 203–206.
- [33] H. Sarimveis, A. Nikolakopoulos, A line up evolutionary algorithm for solving nonlinear constrained optimization problems, *Computers and Operations Research* 32 (2005) 1499–1514.
- [34] R.L. Becterra, C.A.C. Coello, Cultured differential evolution for constrained optimization, *Computer Methods in Applied Mechanics and Engineering* 195 (2006) 4303–4322.
- [35] M.J.D. Powell, A Direct search optimization method that models the objective and constraint functions by linear interpolation, DAMTP/NA5, Cambridge, England.
- [36] I.G. Tsoulos, Modifications of real code genetic algorithm for global optimization, *Applied Mathematics and Computation* 203 (2008) 598–607.
- [37] A.V. Levy, A. Montalvo, The tunneling algorithm for global optimization of functions, *SIAM Journal of Scientific and Statistical Computing* 6 (1985) 15–29.
- [38] H.M. Salkin, *Integer Programming*, Edison Wesley Publishing Com., Amsterdam, 1975.
- [39] D.M. Himmelblau, *Applied Nonlinear Programming*, McGraw-Hill, New York, 1972.
- [40] R. Hess, A heuristic search for estimating a global solution of non convex programming problems, *Operations Research* 21 (1973) 1267–1280.
- [41] K. Schittkowski, More examples for mathematical programming codes, *Lecture Notes in Economics and Mathematical Systems* (1987) 282.
- [42] P. Chootinan, A. Chen, Constraint handling in genetic algorithms using a gradient – based repair method, *Computer and Operations Research* 33 (2006) 2263–2281.
- [43] Y.C. Lin, K.S. Hwang, F.S. Wang, Hybrid differential evolution with multiplier updating method for nonlinear constrained optimization problems, in: *Proceedings of the 2002 IEEE World Congress on Computation Intelligence*, IEEE Service Center, Honolulu Hawaii, 2002, pp. 872–877.
- [44] J.J. Liang, T.P. Runarsson, E. Mezura-Montes, M. Clerc, P.N. Suganthan, C.A.C. Coello, K. Deb, Problem definitions and evaluation criteria for the CEC2006 special session on constrained real-parameter optimization. <http://www.ntu.edu.sg/home/EPNSugan/index_files/CEC-06/CEC06.htm>.
- [45] C.A.C. Coello, Use of a self-adaptive penalty approach for engineering optimization problems, *Computers and Industrial Engineering* 41 (2000) 113–127.
- [46] P. Spelluci, An SQP method for general nonlinear programs using only equality constrained subproblems, *Mathematical Programming* 82 (1998) 413–448.
- [47] P. Spelluci, A new technique for inconsistent problems in the SQP method, *Mathematical Methods of Operational Research* 47 (1998) 355–400.